BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 2 . MODULE 3 OF 16

Module 3: Tissues

Topics: Epithelial Tissue Classification, Connective Tissues, Muscle and Nervous Tissue Overview

TOPIC 1 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Epithelial Tissue Classification

Naming epithelia by layer and cell shape; where each type lives and what it does.

The Science

Epithelia cover and line. They are everywhere there is a body surface, external (skin) or internal (gut lumen, blood vessel, alveolus). Their job is barrier, exchange, secretion, or absorption, depending on location. They are avascular; nutrients come from the underlying connective tissue.

Classification is by layer count (first name) and cell shape (second name). Simple = one layer, made for exchange or secretion. Stratified = multiple layers, made for protection. Squamous = flat (good for diffusion). Cuboidal = cubed (room for secretion machinery). Columnar = tall (room for both, with microvilli for absorption). Once they have the naming logic, they can predict structure from function and vice versa.

Special cases: pseudostratified columnar (looks layered but is not, lines the upper airway, has cilia) and transitional (urinary bladder, can stretch flat). Spend extra time on the upper airway: pseudostratified ciliated columnar with goblet cells is the mucociliary escalator. When that fails (smoking, cystic fibrosis), the lungs lose their main defense. Predicting structure from function is the highest-yield exam skill in this section.

Teaching This Topic

Before the video (drawing prompt)

Show students a list of locations (alveolus, esophagus, kidney tubule, bladder, upper airway, small intestine) and have them predict the epithelial type for each before watching.

Common misconceptions to address

- Pseudostratified is not actually stratified. Every cell still touches the basement membrane.
- Transitional epithelium does not appear transitional under a microscope; the name means it transitions in appearance when the bladder fills.
- Students confuse 'simple' for 'simple to understand.' It means one cell layer.

Order of operations (what to teach first)

Function categories (barrier, exchange, secretion), then the naming system, then examples by location, then the special cases.

Self-test prompts

- Name the epithelium of an alveolus and explain why.
- Why is the upper airway pseudostratified with cilia rather than stratified squamous?
- What changes about transitional epithelium between an empty and full bladder?

Why This Matters to Your Patients

Chronic smokers undergo squamous metaplasia of the upper airway: protective stratified squamous replaces the original ciliated columnar. The new tissue cannot clear mucus, and pathogens linger. This is the cellular start of COPD. Burns destroy epidermis (stratified squamous) and demand grafting because the protective barrier is the entire point of skin. Barrett's esophagus replaces esophageal stratified squamous with columnar (in response to acid reflux); the new tissue is metabolically more active and more likely to become cancerous. Almost every tissue-level diagnosis in pathology is an epithelial story.

TOPIC 2 OF 3 . WEEK 2 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Connective Tissues

Cells embedded in a matrix: loose, dense, cartilage, bone, blood.

The Science

Connective tissue is defined by what is between the cells, not the cells themselves. Cells suspended in an extracellular matrix of ground substance plus protein fibers (collagen, elastic, reticular). Vary the matrix and you get loose connective tissue, dense connective tissue, cartilage, bone, or blood. One family, enormous range of function.

Loose connective (areolar, adipose, reticular) is the general-purpose packing. Dense regular (tendons, ligaments) has parallel collagen for tensile strength in one direction. Dense irregular (dermis) has interwoven collagen for strength in all directions. Cartilage has chondrocytes in lacunae within a flexible matrix; it is avascular and heals poorly. Bone has osteocytes in a rigid mineralized matrix. Blood is connective tissue with a liquid matrix (plasma).

Repair maps onto vascularity. Loose connective heals fast (rich blood supply). Cartilage heals poorly (avascular; nutrients arrive by diffusion only). Bone heals well, contrary to intuition, because it is highly vascular and has active osteoblasts ready to lay down new matrix. This is why a torn meniscus often fails to heal but a broken femur usually does.

Teaching This Topic

Before the video (drawing prompt)

Ask students to list five connective tissues and rank them by how well they heal. They have to predict before the data.

Common misconceptions to address

- Blood is connective tissue. This is counterintuitive but high-yield.
- Adipose is sometimes seen as inert padding. It is metabolically and endocrinologically active.
- Students assume cartilage and bone are similar because they are both 'hard.' Their vascular biology and repair capacity are opposite.

Order of operations (what to teach first)

What defines connective tissue (matrix), then types organized by matrix composition, then repair, organized by vascularity.

Self-test prompts

- Without notes, list five connective tissue types.
- Why does cartilage heal poorly while bone heals well?
- Why is blood classified as connective tissue?

Why This Matters to Your Patients

Osteogenesis imperfecta (defective collagen) makes bones brittle but also affects tendons, ligaments, and the sclera. Ehlers-Danlos syndromes affect collagen and produce hypermobile joints and fragile skin. A meniscal tear often requires surgery because cartilage cannot self-repair; a torn ACL has limited repair capacity for the same reason. Tendinopathy is a slow disease because the blood supply to tendons is limited. Burn patients lose dermis and require grafting because the dense irregular collagen network underneath the epidermis is the structural foundation skin needs.

TOPIC 3 OF 3 . WEEK 2 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Muscle and Nervous Tissue Overview

Three muscle types compared, plus neurons and glia at a tissue level.

The Science

Three muscle types, two nervous tissue cells: a small set, but high-yield. Compare-and-contrast pedagogy works best here.

Skeletal muscle: striated, multinucleated, voluntary, attached to bone. Cardiac: striated, single nucleus, involuntary, has intercalated discs (gap junctions that allow the heart to act as a functional syncytium). Smooth: not striated, single nucleus, involuntary, walls of hollow organs (gut, vessels, bladder, uterus). The discriminating features are striation (yes for skeletal and cardiac, no for smooth) and nucleus count (many for skeletal, one for the others).

Nervous tissue is neurons plus glia. Neurons are the excitable, signaling cells. Glia are the support cells: in the CNS, astrocytes (blood-brain barrier and metabolism), oligodendrocytes (myelin), microglia (immune), ependymal (CSF production). In the PNS, Schwann cells make myelin and satellite cells support cell bodies. Glia outnumber neurons and do most of the day-to-day maintenance. When myelin fails (multiple sclerosis in CNS, Guillain-Barre in PNS), conduction fails.

Teaching This Topic

Before the video (drawing prompt)

Have students predict, for each muscle type: striated yes or no, nuclei one or many, voluntary or not, where in the body.

Common misconceptions to address

- Cardiac muscle is sometimes lumped with skeletal because both are striated. The regulation and conduction biology are very different.
- Smooth muscle is thought of as weak. It is enormously strong; the uterus in labor and the GI tract are smooth muscle.
- Glia are thought of as background. They do most of the work of the nervous system.

Order of operations (what to teach first)

Three muscle types side-by-side (compare on a small set of features), then neurons as excitable cells, then glia as the support network.

Self-test prompts

- Which muscle type has intercalated discs and why does that matter?
- What CNS cell makes myelin?
- What goes wrong physiologically when myelin is destroyed?

Why This Matters to Your Patients

Multiple sclerosis destroys CNS myelin and produces conduction failure across whatever tract is affected: vision changes if the optic nerve is hit, weakness or numbness if spinal tracts are hit, coordination loss if cerebellar pathways are hit. Guillain-Barre syndrome attacks PNS myelin and produces ascending paralysis. Smooth muscle disorders include irritable bowel syndrome (dysregulated gut motility), urinary incontinence (detrusor dysfunction), and asthma (bronchial smooth muscle hyperreactivity). Cardiac myocyte loss is the substrate of heart failure. Three muscle types, three families of disease.